IACR Communications in Cryptology
ISSN 3006-5496, Vol. 3, No. 1, 23 pages.

https://doi.org/10.62056/avivr-zn4

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More Efficient Lattice-based OLE

from Circuit-private Linear HE with Polynomial Overhead

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Abstract. We present a new and efficient method to obtain circuit privacy for lattice-based linearly homomorphic encryptions (LHE). In particular, our method does not involve noise-flooding with superpolynomially large errors or iterative bootstrapping. As a direct result, we obtain a semi-honest oblivious linear evaluation (OLE) protocol with the same efficiency, reducing the communication cost of the prior state of the art by 50%. Consequently, the amortized time of our protocol improves upon the prior work by 33% under a 100Mbps network setting. Our semi-honest OLE is the first to achieve both concrete efficiency and asymptotic quasi-optimality.

Keywords: Homomorphic encryption · Circuit privacy · Oblivious linear evaluation · Secure multi-party computation

1 Introduction

Homomorphic encryption (HE) [RAD78, Gen09], which is a cryptosystem that supports computation on encrypted data, is a versatile tool in cryptography. This is still true even when homomorphic computation of *linear functions* is supported. So-called *linearly homomorphic encryption* (LHE) has found use in various contexts [CGS97, OS07, BDOZ11, KPR18, dCJV21].

However, when designing cryptographic protocols with HE, we often need to guarantee not only data privacy but also *circuit privacy*. That is, we want the resulting ciphertext of homomorphic evaluation not to leak any information about the evaluated circuit other than the output of the circuit. For lattice-based HE schemes, several methods to achieve circuit-privacy have been proposed, but they are unsatisfactory in terms of practical efficiency. They either (i) use the *noise-flooding* technique that adds *superpolynomially large* noise to the resulting ciphertexts [Gen09], or (ii) repeat exhaustive fully homomorphic encryption (FHE) *bootstrapping* [DS16].[1]

In this work, we introduce a new and efficient technique to achieve circuit-private LHE from the BFV scheme [Bra12, FV12]. Our technique does not involve noise-flooding or FHE bootstrapping, allowing us to maintain low computation costs while our ciphertext modulus is only polynomial in the security parameter. To demonstrate the performance, we apply our efficient LHE to construct an oblivious linear evaluation (OLE) protocol.

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[1] One exception is [BdPMW16]. See Sec. 1.3.

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Received: 2026-02-03 Accepted: 2026-04-21

Oblivious linear evaluation (OLE) [NP99] is an arithmetic analogue of oblivious transfer. OLE is a two-party protocol between a sender with $a, b \in R$ and a receiver with $x \in R$, where R is some finite ring. At the end of the protocol, the receiver obtains the value of $ax+b \in R$ while learning nothing about a and b, and the sender learns nothing about x. OLE is a fundamental building block in various cryptographic protocols: general-purpose secure multi-party computation (MPC) [IPS09, ADI+17, DGN+17, HIMV19, CDI+19, EGP+23], zero-knowledge proof (ZKP) [BCGI18, WYKW21, YSWW21], and private set intersection (PSI) [GN19, RS21, CILO22].

A circuit-private LHE can directly lead to a two-round OLE protocol with passive security. This new OLE protocol inherits the compact HE parameters from the LHE scheme, resulting in lower communication costs than the noise-flooding approach [dCJV21].

Furthermore, our semi-honest OLE is asymptotically quasi-optimal (AQO)[2] [dCHI+22]. Our protocol is concretely much more efficient and arguably more intuitive than the previous AQO-OLE of [dCHI+22], which relies on *correlation extractor* of [BGMN18].

1.1 Our Contribution

In this work, we construct a circuit-private linear HE scheme from the RLWE-based BFV scheme [Bra12, FV12]. The existing method relies on either bootstrapping [DS16] or noise flooding [Gen09], which introduces significant overhead in terms of space or time complexity. Our method is based on a simple observation that there is still room for further randomizations in the conventional homomorphic evaluation algorithm beyond noise flooding. More precisely, we observe that the coefficients of a linear function in the plaintext space have several corresponding elements in the ciphertext space, such that any of them can be used in the evaluation while preserving the correctness of homomorphic computation. Our algorithm is surprisingly simple in that it only requires one more Gaussian sampling and a single encryption compared to the conventional (non-circuit-private) linear evaluation algorithm for BFV. Furthermore, its overhead in parameter size is very small, in terms of both concrete and asymptotic manners. We also extend this idea to design an efficient semi-honest OLE protocol.

1.2 Technical Overview

Let $R := \mathbb{Z}[X]/(X^n+1)$ be a polynomial ring, and $R_q := R/qR$ and $R_t := R/tR$ be residue rings where q and t are ciphertext and plaintext moduli such that $t \mid q$, and n is some power-of-two. Then, given a public key $\mathsf{pk} = (p_0, p_1) \in R_q^2$ such that $p_0 + p_1 s = e_p \bmod q$ for the secret s and noise e_p, the BFV encryption of a message $x \in R_t$ is given by $\times$ $c := e_2 \cdot \mathsf{pk} + (e_0 + \Delta x, e_1) \bmod q$ for $\Delta = q/t$ and some small ring elements e_0, e_1 and e_2. For simplicity, we write this as $c := \mathtt{Enc}(x, e = (e_0, e_1, e_2))$. Now let us consider an affine function $f(z) = az + b$ over R_t for some coefficients $a, b \in R_t$. In BFV, the conventional way of homomorphically evaluating the linear function f is to compute $\mathbf{c}' = r \cdot \mathbf{c} + (\Delta b, 0) \pmod q$ where $r = [a]_t \in R$.

However, this method does not guarantee circuit privacy since the second component of $\mathbf{c}'$ completely reveals the information about $r = [a]_t$, and therefore b is also fully recovered. To achieve circuit privacy, prior works adopted a method to add an encryption of zero with an exponentially large randomness $\mathbf{e}'$ to the resulting ciphertext $\mathbf{c}'$, so-called noise-flooding, so that the information of a and b are statistically obliterated. However, this method yields an exponentially large ciphertext modulus, which significantly increases computation and communication costs.

This work starts with the following observations: (i) r does not necessarily need to be exactly $[a]_t$, but can be any small element in the coset $a + t\mathbb{Z}^n$ to guarantee correctness,

[2] We say a cryptographic scheme is *asymptotically quasi-optimal (AQO)* if it solves a size-n cryptographic problem with $\tilde{O}(n + \lambda)$ cost, where λ is the security parameter.

and (ii) the addition of two discrete Gaussians over the coset $a + t\mathbb{Z}^n$ and $\mathbb{Z}^n$ with a proper width parameter can be statistically close to a discrete Gaussian distribution over $\mathbb{Z}^n$, which is notably independent of a. From these observations, we construct a new linear evaluation algorithm as follows. We first sample r and $\mathbf{e}'$ from discrete Gaussian distributions $D_{a+t\mathbb{Z}^n,\sigma}$ and $D^3_{\mathbb{Z}^n,\tau}$ respectively, for some width parameters $\sigma, \tau > 0$. Then, we compute a ciphertext $\mathbf{c}'$ as

$$\mathbf{c}' := r \cdot \mathbf{c} + \mathtt{Enc}(b, \mathbf{e}') \pmod q.$$

Interestingly, our algorithm computes an essentially identical formula to the previous method; however, the difference in the sampling procedure of r results in a significant reduction in the size of $\mathbf{e}'$. Technically, by setting $\sigma = O(t)$ and $\tau = O(\sigma n \|\mathbf{e}\|_\infty)$, the distribution of the convolved noise term $r \cdot \mathbf{e} + \mathbf{e}'$ of $\mathbf{c}'$ is statistically indistinguishable from the discrete Gaussian distribution $D_{\mathbb{Z}^{3n},\boldsymbol{\kappa},\sqrt{\Sigma}}$, where both $\boldsymbol{\kappa}$ and Σ are independent of r (refer to Theorem 1). This fact directly implies that the BFV scheme with our linear evaluation algorithm is circuit private, since $\mathbf{c}' = r \cdot \mathtt{Enc}(x, \mathbf{e}) + \mathtt{Enc}(b, \mathbf{e}') = \mathtt{Enc}(ax + b, r \cdot \mathbf{e} + \mathbf{e}') \pmod q$ and the only information dependent on a and b in $\mathbf{c}'$ is the resulting message $ax + b$.

When we compare the correctness condition of our linear evaluation algorithm (Lem. 5) to the non-circuit-private conventional algorithm, we can easily verify that our algorithm only requires $\tilde{O}_\lambda(1)$ additional bits on the ciphertext modulus q for the width parameters $\sigma = O(t)$ and $\tau = O(\sigma n \|\mathbf{e}\|_\infty)$. Therefore, our linear evaluation algorithm provides circuit privacy *almost for free*.

Finally, we construct a semi-honest OLE protocol based on our circuit-private LHE. Note that this can be directly obtained from a homomorphic linear evaluation algorithm with circuit privacy. To be precise, the receiver first sends an encryption $\mathbf{c}$ of the plaintext x, and the sender evaluates an affine function $ax + b$ on the input ciphertext with the circuit private LHE and sends back the output encryption $\mathbf{c}'$ of $ax + b$. Then, the output ciphertext does not include any information about the coefficients a and b. As a result, we could reduce the communication cost of the OLE protocol compared to the state-of-the-art semi-honest OLE protocol based on linear HE [dCJV21].

1.3 Related Work

Circuit Privacy. While most approaches to achieving circuit privacy for lattice-based HE leverage noise-flooding [Gen09] or iterative FHE bootstrapping [DS16], Bourse-DelPino-Minelli-Wee [BdPMW16] presented a new and efficient method to achieve circuit privacy through *randomized gadget decomposition*. However, their approach only applies to GSW-style encryptions [GSW13]. We note that GSW-style encryptions suffer from significantly larger ciphertext sizes and higher computation costs compared to BFV encryption, hindering practical usage.

Oblivious Linear Evaluation. Several prior works [BEPU+20, dCJV21] have attempted to optimize OLE protocols while still relying on the noise-flooding technique. The work of Baum et al. [BEPU+20] constructs a one-round (simultaneous message) protocol for generating *random* OLE correlations. However, the ciphertexts in this protocol require either one or two applications of noise flooding, resulting in high communication costs. The work of de Castro, Juvekar, and Vaikuntanathan [dCJV21] optimized the "folklore" noise-flooding approach by removing the need to sample the flooding noise. While this saves computation time, the communication remains identical to that of the noise-flooding approach.

As a point of theoretical interest, the work of de Castro et al. [dCHI+22] constructed AQO-OLE protocols. We observe that our semi-honest OLE protocol also achieves the

same asymptotic performance without requiring the computationally expensive machinery of correlation extractors [BGMN18] as in the previous work.

A recent category of the OLE protocols generates random correlations with communication that is asymptotically sublinear in the length of the correlations [BCG+20, CRR21]. For the task of generating *random* OLE correlations, these protocols will eventually outperform our work. However, even beyond this crossover point, these protocols require a substantial amount of *seed* OLEs to expand to the full protocol output. Therefore, our protocol remains applicable even in the setting where the overall sublinear correlation generator outperforms any linear-time OLE protocol.

There are previous works [BDM22, BBDP22] that achieve OLE protocols with appealing properties. For instance, [BBDP22] achieves *rate-1* OLE. However, they are mostly theoretical results, and an analysis of their concrete efficiency is not provided. On the other hand, we aim to achieve a concretely efficient OLE protocol.

1.4 Organization

The rest of the paper is organized as follows. Section 2 reviews some preliminaries on lattices and RLWE and describes OLE-related functionalities. In Section 3, we present a new idea to construct a circuit-private HE scheme for affine functions over a polynomial ring. In Section 4, we present efficient OLE protocols against passive adversaries, along with correctness proofs and security analysis.

2 Preliminary

2.1 Basic Notations and Terminology

For two distributions $\mathcal{D}_1$ and $\mathcal{D}_2$ over a countable domain Ω, the *statistical distance* between $\mathcal{D}_1$ and $\mathcal{D}_2$ is defined as $\frac{1}{2}\sum_{x\in\Omega}|\mathcal{D}_1(x)-\mathcal{D}_2(x)|$. For a distribution $\mathcal{D}$ over Ω, we denote by $x \leftarrow \mathcal{D}$ the sampling x from $\mathcal{D}$. When $\mathcal{D}$ is the uniform distribution over Ω, we simply denote the sampling by $x \leftarrow \Omega$. For two vectors u, v of the same size, we denote the point-wise multiplication (Hadamard multiplication) by $u \odot v$.

Let $R := \mathbb{Z}[X]/(X^n+1)$ be the $(2n)$-th cyclotomic ring and $R_q := \mathbb{Z}_q[X]/(X^n+1)$ be the residue ring of R modulo an integer q. We identify $a = \sum_{i=0}^{n-1} a_i X^i$ in R (or R_q) with the vector of its coefficients $\mathbf{a} = (a_0, \cdots, a_{n-1})$. Hence, $\|a\|_\infty$ and $\|a\|_1$ denote the infinity norm and ℓ^1-norm of $\mathbf{a}$, respectively. We also identify a ring element $a = \sum_{i=0}^{n-1} a_i X^i \in R$ with a negacyclic matrix $\mathbf{A}$, where

$$\mathbf{A} = \begin{bmatrix} a_0 & -a_{n-1} & \dots & -a_1 \\ a_1 & a_0 & \dots & -a_2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1} & a_{n-2} & \dots & a_0 \end{bmatrix}.$$

For any $a, b \in R$, it holds that $\|ab\|_\infty \le \|a\|_\infty \|b\|_1 \le n\|a\|_\infty\|b\|_\infty$. Moreover, the vector representation of ab is obtained as $\mathbf{Ab}$, where $\mathbf{A}$ is the negacyclic matrix of a, and $\mathbf{b}$ is the coefficient vector of b.

A symmetric real matrix $\mathbf{\Sigma} \in \mathbb{R}^{n\times n}$ is called *positive-definite*, if $\mathbf{x}^\top\mathbf{\Sigma}\mathbf{x} > 0$ holds for all $\mathbf{0} \neq \mathbf{x} \in \mathbb{R}^n$. Equivalently, positive-definite matrices can be characterized by their spectral decomposition of the form $\mathbf{\Sigma} = \mathbf{Q}\mathbf{D}^2\mathbf{Q}^\top$, where $\mathbf{Q} \in \mathbb{R}^{n\times n}$ is an orthogonal matrix (i.e., $\mathbf{Q}^{-1} = \mathbf{Q}^\top$) and $\mathbf{D}$ is a diagonal matrix with positive diagonal entries. Thus, any positive-definite $\mathbf{\Sigma}$ is invertible, and $\mathbf{\Sigma}^{-1}$ is also positive-definite.

We say that matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is a square root of $\mathbf{\Sigma} \in \mathbb{R}^{n\times n}$, written $\mathbf{A} = \sqrt{\mathbf{\Sigma}}$, if $\mathbf{A}\mathbf{A}^\top = \mathbf{\Sigma}$ holds. We use the notation $\sqrt{\mathbf{\Sigma}}$ when the specific choice of square root is

irrelevant. Note that every positive-definite matrix has a square root by the above spectral decomposition.

For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, we define the *matrix norm* of $\mathbf{A}$ as:

$$\|\mathbf{A}\| = \sup_{\mathbf{0} \neq \mathbf{x} \in \mathbb{R}^n} \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2}.$$

We denote by $s_{\max}(\mathbf{A})$ and $s_{\min}(\mathbf{A})$ the largest and smallest singular values of $\mathbf{A}$, respectively. Note that $\|\mathbf{A}\| = s_{\max}(\mathbf{A})$.

$[a]_t$ denotes the residue of a modulo t, *i.e.,* $[a]_t = a \pmod{t}$.

2.2 Discrete Gaussians on Lattices

The n-dimensional Gaussian function $\rho : \mathbb{R}^n \mapsto (0, 1]$ is defined as

$$\rho(\mathbf{x}) = \exp(-\pi \cdot \|\mathbf{x}\|_2^2).$$

For a non-singular matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and a vector $\boldsymbol{\mu} \in \mathbb{R}^n$, we define $\rho_{\boldsymbol{\mu},\mathbf{A}}(\mathbf{x})$ as $\rho\left(\mathbf{A}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$. Note that for a positive-definite $\boldsymbol{\Sigma}$, we have

$$\rho_{\boldsymbol{\mu},\sqrt{\boldsymbol{\Sigma}}}(\mathbf{x}) = \exp\left(-\pi(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right).$$

When $\boldsymbol{\mu} = 0$, we omit $\boldsymbol{\mu}$ and denote it by $\rho_{\sqrt{\boldsymbol{\Sigma}}}(\mathbf{x})$, for simplicity. If $\boldsymbol{\Sigma}$ is also $\sigma^2\mathbf{I}$ for some $\sigma > 0$, we denote it by $\rho_\sigma(\mathbf{x})$.

A *lattice* Λ is a discrete additive subgroup of $\mathbb{R}^n$. The *dual lattice* of a lattice $\Lambda \subset \mathbb{R}^n$ is defined as $\Lambda^* = \{\mathbf{v} \in \mathbb{R}^n \mid \forall \mathbf{u} \in \Lambda, \langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{Z}\}$. For a lattice $\Lambda \subset \mathbb{R}^n$, a vector $\boldsymbol{\mu} \in \mathbb{R}^n$, and a positive-definite matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$, we define the *discrete Gaussian distribution* over Λ, centered at $\boldsymbol{\mu}$, and with covariance $\boldsymbol{\Sigma}$ as

$$D_{\Lambda,\boldsymbol{\mu},\sqrt{\boldsymbol{\Sigma}}}(\mathbf{x}) = \frac{\rho_{\boldsymbol{\mu},\sqrt{\boldsymbol{\Sigma}}}(\mathbf{x})}{\rho_{\boldsymbol{\mu},\sqrt{\boldsymbol{\Sigma}}}(\Lambda)}.$$

It is simply denoted by $D_{\Lambda,\boldsymbol{\mu},\sigma}(\mathbf{x})$ when $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$ for some $\sigma > 0$. When $\boldsymbol{\mu} = 0$, we omit $\boldsymbol{\mu}$ and denote it by $D_{\Lambda,\sqrt{\boldsymbol{\Sigma}}}(\mathbf{x})$ (or $D_{\Lambda,\sigma}(\mathbf{x})$).

Definition 1 (Smoothing parameter [MR07])**.** For an n-dimensional lattice Λ and positive real $\epsilon > 0$, the smoothing parameter $\eta_\epsilon(\Lambda)$ is the smallest s such that $\rho_{1/s}(\Lambda^* \backslash \{\mathbf{0}\}) \leq \epsilon$.

Lemma 1 ([GPV08, Lem. 4.2])**.** *For any $c \in \mathbb{R}$, any $\epsilon > 0$, any $\sigma \geq \eta_\epsilon(\mathbb{Z})$, and any $\kappa > 0$, the following inequality holds.*

$$\Pr_{x \leftarrow D_{c+\mathbb{Z},\sigma}}[|x - c| \geq \kappa \cdot \sigma] \leq 2e^{-\pi\kappa^2} \cdot \frac{1+\epsilon}{1-\epsilon}$$

The following lemma states that the total Gaussian measures over cosets of a given lattice are essentially the same when the covariance is *sufficiently large* compared to the smoothing parameter.

Lemma 2 (Simplified Convolution Lemma [Pei10])**.** *Let $\sigma_1, \sigma_2 > 0$ be reals such that $\sigma_3^{-2} := \sigma_1^{-2} + \sigma_2^{-2}$ satisfies $\sigma_3 \geq \eta_\epsilon(\mathbb{Z}^n)$ for some $0 < \epsilon < 1/2$. Then for an arbitrary $\mathbf{c} \in \mathbb{Z}^n$, the distribution*

$$\{\mathbf{x}_1 + \mathbf{x}_2 \mid \mathbf{x}_1 \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma_1},\ \mathbf{x}_2 \leftarrow \mathcal{D}_{\mathbb{Z}^n,\mathbf{c},\sigma_2}\}$$

is within statistical distance 2ϵ of $\mathcal{D}_{\mathbb{Z}^n,\mathbf{c},\sqrt{\sigma_1^2+\sigma_2^2}}$.

Lemma 3 ([AGHS13, Lem. 4])**.** *Let Λ be a full-rank n-dimensional lattice. For any real $\epsilon \in (0,1)$, vector $\mathbf{c} \in \mathbb{R}^n$, and non-singular matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ such that $s_{\min}(\mathbf{A}) \geq \eta_\epsilon(\Lambda)$,*

$$\rho_{\mathbf{A}}(\Lambda + \mathbf{c}) \in \left[\frac{1-\epsilon}{1+\epsilon}, 1\right] \cdot \rho_{\mathbf{A}}(\Lambda).$$

Finally, we note a useful bound on the smoothing parameters.

Lemma 4 ([MR07, Lem. 3.3])**.** *For any n-dimensional lattice Λ and $\epsilon > 0$,*

$$\eta_\epsilon(\Lambda) \leq \sqrt{\frac{\ln(2n(1+1/\epsilon))}{\pi}} \cdot \lambda_n(\Lambda)$$

where $\lambda_n(\Lambda)$ is the n-th successive minimum of Λ. That is, $\lambda_n(\Lambda)$ is the smallest real number $r > 0$ such that $\dim(\mathrm{span}(\Lambda \cap rB)) = n$ where B is the n-dimensional unit ball centered at the origin.

The following is a direct corollary of Lem. 4.

Corollary 1. *For any $t > 0$ and $\epsilon > 0$, the following inequality holds.*

$$\eta_\epsilon(t\mathbb{Z}^n) \leq t \cdot \sqrt{\frac{\ln(2n(1+1/\epsilon))}{\pi}}$$

2.3 RLWE

Definition 2. Let n be a power of two, $q > 0$ be an integer and $\rho > 0$ be a real number. The Ring Learning with Errors (RLWE) distribution with parameter (n, q, ρ) and a secret $s \in R$ is a distribution over R_q^2 obtained by sampling $a \leftarrow R_q$ and $e \leftarrow D_{\mathbb{Z}^n,\rho}$, then returning (b, a) where $b = as + e \pmod q$. For a secret key distribution χ over R, the decisional RLWE problem denoted by $\mathsf{RLWE}(n, q, \rho, \chi)$ is to distinguish the RLWE distribution for a secret $s \leftarrow \chi$ from a uniform distribution over R_q^2.

Throughout the paper, we assume that the secret is chosen from the ternary distribution, which means selecting each coefficient from $\{-1, 0, 1\}$. When χ is a discrete Gaussian distribution with width parameter σ, we simply denote the RLWE problem by $\mathsf{RLWE}(n, q, \rho, \sigma)$.

2.4 Oblivious Linear Evaluation

The oblivious linear evaluation (OLE) is a secure computation protocol for a linear function between two parties: the receiver and the sender. A standard OLE over a field F allows the sender with $a, b \in F$ to securely compute $ax + b$ and send the output to the receiver who holds $x \in F$ [NP99, IPS09]. As an extension of the conventional OLE protocol, there exist two variants of the OLE protocol, namely *Batch OLE* (BOLE) and *Vector OLE* (VOLE). In BOLE, the receiver can learn a set of values $a_i \cdot x_i + b_i$, where the sender holds $a_i, b_i \in F$ and the receiver holds $x_i \in F$ for $1 \leq i \leq n$. To put it in other words, BOLE enables the receiver to perform multiple oblivious linear evaluations through a single protocol. In VOLE, the receiver learns a vector $\mathbf{a}x + \mathbf{b}$ where $\mathbf{a}, \mathbf{b} \in F^n$ and $x \in F$. VOLE can be viewed as a special case of a BOLE protocol in which the receiver sends only one input x.

More generally, the standard OLE functionality can be extended over a (finite) ring, referred to as *ring-OLE* [CILO22]. Throughout the paper, we concentrate on the ring-OLE protocol over the residue ring R_t, since it is amenable to parallelism for VOLE and BOLE (when the ring dimension n and the plaintext modulus t satisfy the condition $2n \mid t - 1$). The formal descriptions of the functionalities of key generation and OLE are given in Fig. 1 and Fig. 2, respectively.

Functionality $\mathcal{F}_{\mathsf{KeyGen}}$

Input: a public parameter $pp = (n, q, \rho)$ and a secret key distribution χ.
Output:

1. Sample a secret key $s \leftarrow \chi$. Sample $a \leftarrow R_q$ and $e \leftarrow D_{\mathbb{Z}^n,\rho}$, and set the public key as $\mathbf{p} = (-as + e, a) \in R_q^2$.
2. Send s and $\mathbf{p}$ to the receiver.
3. Send $\mathbf{p}$ to the sender.

Figure 1: The KeyGen functionality.

Functionality $\mathcal{F}_{\mathsf{OLE}}$

Input:

1. A message $x \in R_t$ from the receiver
2. Coefficients $a, b \in R_t$ of a linear function $f(z) = az + b$ from the sender.

Output:

1. Send $f(x) = ax + b \in R_t$ to the receiver.
2. The sender receives nothing.

Figure 2: The OLE functionality.

3 Circuit-private Linear Evaluation from BFV

In this section, we present a novel idea to build a circuit-private HE scheme for affine functions over a polynomial ring. Our approach is quite simple but offers a randomized evaluation method with only a small overhead compared to the naïve approach.

We start by describing the RLWE-based BFV scheme as a *linearly homomorphic encryption* scheme and present a new linear evaluation algorithm. Then, we provide correctness and security proofs for our algorithm.

3.1 The BFV Scheme

Below, we provide a brief description of the RLWE-based BFV encryption scheme [Bra12, FV12].

• $\texttt{Setup}(1^\lambda)$: Given a security parameter λ, choose a power-of-two integer n, a ciphertext modulus q, a plaintext modulus t, an error parameter $\rho > 0$, and a key distribution χ over R. Return the public parameter $pp = (n, q, t, \rho, \chi)$.

• $\texttt{KeyGen}(pp)$: Sample a secret key $s \leftarrow \chi$. Sample $p_1 \leftarrow R_q$ and $e_p \leftarrow D_{\mathbb{Z}^n,\rho}$, and set a public key as $\mathbf{p} = (p_0, p_1)$ where $p_0 = -p_1 \cdot s + e_p \pmod q$.

• $\texttt{Enc}_{\mathbf{p}}(x)$: For a message $x \in R_t$, sample $\mathbf{e} \leftarrow D^3_{\mathbb{Z}^n,\rho}$, return $\mathbf{c} = \texttt{Enc}_{\mathbf{p}}(x, \mathbf{e})$.

• $\mathtt{Enc}_{\mathbf{p}}(x, \mathbf{e})$: For a message $x \in R_t$ and encryption randomness $\mathbf{e} = (e_0, e_1, e_2) \in R^3$, return $\mathbf{c} = e_2 \cdot \mathbf{p} + (\Delta \cdot x + e_0, e_1) \pmod q$.

• $\mathtt{Dec}_s(\mathbf{c})$: For $\mathbf{c} = (c_0, c_1)$, return $\lfloor (c_0 + c_1 s)/\Delta \rceil \pmod t$.

Throughout this paper, we suppose that the plaintext modulus t divides the ciphertext modulus q. This assumption simplifies the construction of our algorithms and protocols, as well as their security analysis. We provide the security analysis of the general case where t does not divide q in Appendix 3.4. We also denote by $\Delta = q/t$ the scaling factor.

Two BFV encryption algorithms are presented above. By default, the BFV encryption algorithm samples encryption randomness internally, as described in the first algorithm. However, we also exploit the second algorithm in which the randomness for encryption is specified explicitly.

Note that the BFV encryption is semantically secure: $\mathtt{Enc}_{\mathbf{p}}(x)$ is indistinguishable from a uniform random variable over R_q^2 under the RLWE assumption of parameter (n, q, ρ, χ). In addition, a fresh BFV encryption $\mathbf{c} = (c_0, c_1) \in R_q^2$ satisfies that

$$c_0 + c_1 s = e_2 \cdot (p_0 + p_1 s) + (\Delta \cdot x + e_0 + e_1 s) = \Delta \cdot x + e \pmod q$$

where $e = e_0 + e_1 s + e_2 e_p$ denotes the encryption error.

Finally, we describe a linear evaluation algorithm for the BFV scheme. For an affine function $f(z) = az + b$ defined over R_t and a given BFV encryption $\mathbf{c}$ of $x \in R_t$, it homomorphically evaluates f to obtain an encryption of $y = f(x) \in R_t$. However, it cannot be done simply by computing f over the input ciphertext $\mathbf{c} \in R_q^2$ since the coefficients $a, b \in R_t$ cannot be directly used in the computation over the ring R_q. Hence, these coefficients in R_t are embedded into elements of R and R_q via $a \mapsto r := [a]_t \in R$ and $b \mapsto \Delta \cdot b \in R_q$, so that scalar multiplication and addition are well defined over R_q. A formal description is provided below.

• $\mathtt{LinEval}(\mathbf{c}; a, b)$: Given a ciphertext $\mathbf{c} \in R_q^2$ and coefficients $a, b \in R_t$, let $r = [a]_t \in R$ and output $\mathbf{c}' = r \cdot \mathbf{c} + (\Delta b, 0) \pmod q$.

The correctness of this algorithm can be shown as follows: if $\mathbf{c} = \mathtt{Enc}_{\mathbf{p}}(x, \mathbf{e})$ is a BFV encryption for some $x \in R_t$ and $\mathbf{e} \in R^3$, and a, b are elements of R_t with $r = [a]_t$, then we get

$$\mathbf{c}' = r \cdot \mathbf{c} + (\Delta b, 0) = \mathtt{Enc}_{\mathbf{p}}(y, r\mathbf{e}) \pmod q,$$

where $y = rx + b = ax + b \in R_t$.

3.2 Circuit Private Linear Evaluation

As described in the previous section, it is possible to homomorphically evaluate an arbitrary affine function $f(x) = ax + b$ using the BFV scheme; however, the basic linear evaluation algorithm does not guarantee the privacy of the evaluation circuit f. To be precise, for a ciphertext $\mathbf{c} = (c_0, c_1) \in R_q^2$ and $a, b \in R_t$, the linear evaluation algorithm returns $\mathbf{c}' = (c_0', c_1') \leftarrow \mathtt{LinEval}(\mathbf{c}; a, b)$ which satisfies $c_1' = rc_1 \pmod q$ for $r = [a]_t$. Therefore, anyone can recover $a, b \in R_t$ by computing $a = c_1^{-1} \cdot c_1' \pmod t$ and $b = \Delta^{-1} \cdot (\mathbf{c}' - [a]_t \cdot \mathbf{c}) \pmod t$ from a pair of input and output ciphertexts. Similarly, it is also possible to recover r (and thereby $a = r \pmod t$) using the error terms of two ciphertexts: if $c_0 + c_1 s = \Delta \cdot x + e \pmod q$, then $c_0' + c_1' s = \Delta \cdot y + e' \pmod q$ for $e' = re \in R$.

Due to this problem, the notion of *circuit privacy* was introduced [IP07]. Briefly speaking, an evaluation algorithm is said to be circuit-private if there is only negligible information leakage about the evaluation circuit from the input and output ciphertexts. Since this work focuses on linear circuits, the notion of circuit privacy can be formally defined as follows.

Definition 3 (Linear Circuit Privacy)**.** A linearly homomorphic encryption scheme $(\mathtt{Setup}, \mathtt{KeyGen}, \mathtt{Enc}, \mathtt{Dec}, \mathtt{Eval})$ is called circuit private for linear circuits if there exists a PPT simulator algorithm Sim such that for any $x \in R_t$, $a, b \in R_t$, and a PPT algorithm $\mathcal{A}$,

$$|\Pr[\mathcal{A}(\mathtt{Eval}_{\mathsf{pk}}(\mathbf{c}; a, b), \mathbf{c}, \mathsf{sk}, \mathsf{pk})] - \Pr[\mathcal{A}(\mathsf{Sim}(\mathsf{sk}, \mathsf{pk}, \mathbf{c}, y), \mathbf{c}, \mathsf{sk}, \mathsf{pk}) = 1]|$$

is negligible where $pp \leftarrow \mathtt{Setup}(1^\lambda)$, $(\mathsf{sk}, \mathsf{pk}) \leftarrow \mathtt{KeyGen}(pp)$, $\mathbf{c} \leftarrow \mathtt{Enc}_{\mathsf{pk}}(x)$ and $y = ax + b \in R_t$.

To achieve circuit privacy, the *noise-flooding* technique [Gen09] has mostly been used in prior work. This method randomizes a ciphertext by adding an encryption of zero with noise from a wide distribution so that the initial noise is overwhelmed by a newly introduced error. To be specific, a circuit private version of the linear evaluation algorithm can be obtained by modifying the previous evaluation algorithm as $\mathbf{c}' = \mathtt{LinEval}(\mathbf{c}; a, b) + \mathtt{Enc}_{\mathbf{p}}(0, \mathbf{e}')$ where $\mathbf{e}'$ is sampled from a discrete Gaussian distribution with an exponentially large width.

Although the noise flooding technique enables us to achieve circuit privacy, it introduces a significant overhead since the ciphertext modulus q should be exponentially large to support correct decryption of the resulting ciphertext $\mathbf{c}'$, thereby affecting both computation and communication costs.

In this work, we propose a new idea to achieve circuit privacy with much smaller overhead compared to the noise flooding approach. We first remark that the output ciphertext of the previous method can be written as

$$\mathbf{c}' = \mathtt{LinEval}(\mathbf{c}; a, b) + \mathtt{Enc}_{\mathbf{p}}(0, \mathbf{e}') = \mathtt{Enc}_{\mathbf{p}}(y, \tilde{\mathbf{e}}) \pmod q$$

where $r = [a]_t$ and $\tilde{\mathbf{e}} = r \cdot \mathbf{e} + \mathbf{e}' \in R$. Consequently, the noise flooding technique was inevitable to statistically obliterate the information of $r \cdot \mathbf{e}$.

Our main observation is that there is still room for further randomization in the evaluation algorithm. When we embed the coefficient a into an element $r \in R$ before multiplying it by $\mathbf{c}$, an arbitrarily small element of R that is congruent to a modulo t can be chosen instead of setting r exactly as $[a]_t$. The correctness of linear evaluation still holds if r is reasonably small since the plaintext space is R_t. Based on this idea, we propose a novel *randomized* linear evaluation algorithm that samples r from a discrete Gaussian over the coset $a + t\mathbb{Z}^n$, the set of elements $r \in R$ such that $r = a \pmod t$. A formal description is given below:

• $\mathtt{RandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$: Given a ciphertext $\mathbf{c} \in R_q^2$ and ring elements $a, b \in R_t$, sample $r \leftarrow D_{a+t\mathbb{Z}^n, \sigma}$ and $\mathbf{e}' \leftarrow D^3_{\mathbb{Z}^n, \tau}$. Compute and output $\mathbf{c}' := r \cdot \mathbf{c} + \mathtt{Enc}_{\mathbf{p}}(b, \mathbf{e}') \pmod q$.

This surprisingly simple algorithm enables us to use much smaller (asymptotically optimal) parameters and, therefore, achieve substantial performance improvements. In the next section, we will analyze the algorithm and explain how the parameters $\sigma, \tau > 0$ for discrete Gaussian distributions should be chosen to meet the correctness and security requirements.

3.3 Correctness and Security

In this section, we show the correctness and security of our linear evaluation algorithm. We will use B_ρ, B_σ, and B_τ to denote *essential* upper bounds of distributions $D_{\mathbb{Z}^n, \rho}$, $D_{a+t\mathbb{Z}^n, \sigma}$, and $D_{\mathbb{Z}^n, \tau}$, respectively, where a is an arbitrary element in R_t. To put it another way, a sample from each distribution is bounded by the corresponding bound with overwhelming probability. We remark that these bounds can be obtained explicitly from Lem. 1.

Lemma 5 (Correctness)**.** *Let* $\mathbf{c} \leftarrow \mathtt{Enc}_{\mathbf{p}}(x)$ *for some* $x \in R_t$*. Then, the algorithm* $\mathtt{RandLinEval}(\mathbf{c}; a, b)$ *outputs a BFV encryption of* $f(x) = ax + b$ *if*

$$(nB_\rho B_\sigma + B_\tau)(1 + n + nB_\rho) < \Delta/2.$$

Proof. From the definition, we have $\mathbf{c} = \mathtt{Enc}_{\mathbf{p}}(x, \mathbf{e})$ and

$$\mathbf{c}' = r \cdot \mathtt{Enc}_{\mathbf{p}}(x, \mathbf{e}) + \mathtt{Enc}_{\mathbf{p}}(b, \mathbf{e}') = \mathtt{Enc}_{\mathbf{p}}(y, r \cdot \mathbf{e} + \mathbf{e}') \pmod{q} \tag{1}$$

for some $\mathbf{e} \leftarrow D^3_{\mathbb{Z}^n,\rho}$, $r \leftarrow D_{a+t\mathbb{Z}^n,\sigma}$, and $\mathbf{e}' \leftarrow D^3_{\mathbb{Z}^n,\tau}$.

Let $\tilde{\mathbf{e}} = (\tilde{e}_0, \tilde{e}_1, \tilde{e}_2) = r \cdot \mathbf{e} + \mathbf{e}'$. Then, the output ciphertext $\mathbf{c}' = (c'_0, c'_1)$ is equal to $\mathtt{Enc}_{\mathbf{p}}(y, \tilde{\mathbf{e}})$ and satisfies that $c'_0 + c'_1 s = \Delta y + e^* \pmod{q}$ with an error $e^* = \tilde{e}_0 + \tilde{e}_1 s + \tilde{e}_2 e_p$. Note that the key s is sampled from a ternary distribution and each coefficient of $\tilde{\mathbf{e}}$ is bounded by $n \cdot B_\rho B_\sigma + B_\tau$. Therefore, $\|e^*\|_\infty \le (nB_\rho B_\sigma + B_\tau)(1 + n + nB_\rho) < \Delta/2$ and the resulting ciphertext $\mathbf{c}'$ decrypts to $y = ax + b$ correctly. □

We now show that our randomized evaluation algorithm achieves the circuit privacy property. We start with some useful lemmas to analyze the distribution of encryption randomness.

Lemma 6. *Let $t > 0$ be an integer, $\mathbf{E} \in \mathbb{Z}^{m\times n}$ a matrix, and $\sigma, \tau > 0$ reals such that*

$$\frac{1}{\sigma^2} + \frac{1}{\tau^2}\|\mathbf{E}\|^2 \le \frac{1}{\eta_\epsilon(t\mathbb{Z}^n)^2}$$

for some $0 < \epsilon \le 1/2$. Then, for arbitrary $\mathbf{a}$, $\boldsymbol{\mu} \in \mathbb{R}^n$ and $\boldsymbol{\nu} \in \mathbb{R}^m$, the following distribution over $\mathbb{Z}^m$

$$\mathcal{D} := \left\{\mathbf{E}\mathbf{r} + \mathbf{e}' : \mathbf{r} \leftarrow D_{\mathbf{a}+t\mathbb{Z}^n,\boldsymbol{\mu},\sigma}, \mathbf{e}' \leftarrow D_{\mathbb{Z}^m,\boldsymbol{\nu},\tau}\right\}$$

is within statistical distance 4ϵ of $D_{\mathbb{Z}^m,\boldsymbol{\kappa},\sqrt{\boldsymbol{\Sigma}}}$ where $\boldsymbol{\kappa} = \mathbf{E}\boldsymbol{\mu} + \boldsymbol{\nu}$ and $\boldsymbol{\Sigma} = \sigma^2 \cdot \mathbf{E}\mathbf{E}^\top + \tau^2 \cdot \mathbf{I}_m$.

Proof. For $\mathbf{x} \in \mathbb{Z}^m$, the probability that $\mathcal{D}$ outputs $\mathbf{x}$ can be written as follows:

$$\begin{aligned}
\mathcal{D}(\mathbf{x}) &= \Pr[\mathbf{E}\mathbf{r} + \mathbf{e}' = \mathbf{x} \mid \mathbf{r} \leftarrow D_{\mathbf{a}+t\mathbb{Z}^n,\boldsymbol{\mu},\sigma}, \mathbf{e}' \leftarrow D_{\mathbb{Z}^m,\boldsymbol{\nu},\tau}] \\
&= \sum_{\mathbf{y}\in\mathbf{a}+t\mathbb{Z}^n} D_{\mathbf{a}+t\mathbb{Z}^n,\boldsymbol{\mu},\sigma}(\mathbf{y}) \cdot D_{\mathbb{Z}^m,\boldsymbol{\nu},\tau}(\mathbf{x} - \mathbf{E}\mathbf{y}) \\
&\propto \sum_{\mathbf{y}\in\mathbf{a}+t\mathbb{Z}^n} \rho_\sigma(\mathbf{y} - \boldsymbol{\mu}) \cdot \rho_\tau(\mathbf{x} - \boldsymbol{\nu} - \mathbf{E}\mathbf{y}) \\
&= \sum_{\mathbf{y}\in\mathbf{a}-\boldsymbol{\mu}+t\mathbb{Z}^n} \exp\left[-\pi\left(\frac{1}{\sigma^2}\|\mathbf{y}\|^2 + \frac{1}{\tau^2}\|(\mathbf{x} - \boldsymbol{\kappa}) - \mathbf{E}\mathbf{y}\|^2\right)\right].
\end{aligned}$$

Then, we have

$$\begin{aligned}
&\frac{1}{\sigma^2}\|\mathbf{y}\|^2 + \frac{1}{\tau^2}\|(\mathbf{x} - \boldsymbol{\kappa}) - \mathbf{E}\mathbf{y}\|^2 \\
&= \mathbf{y}^\top \boldsymbol{\Sigma}_1^{-1}\mathbf{y} - \frac{1}{\tau^2}\left(\mathbf{y}^\top\mathbf{E}^\top(\mathbf{x} - \boldsymbol{\kappa}) + (\mathbf{x} - \boldsymbol{\kappa})^\top\mathbf{E}\mathbf{y}\right) + \frac{1}{\tau^2}(\mathbf{x} - \boldsymbol{\kappa})^\top(\mathbf{x} - \boldsymbol{\kappa}) \\
&= \left(\mathbf{y} - \frac{1}{\tau^2}\boldsymbol{\Sigma}_1\mathbf{E}^\top(\mathbf{x} - \boldsymbol{\kappa})\right)^\top \boldsymbol{\Sigma}_1^{-1}\left(\mathbf{y} - \frac{1}{\tau^2}\boldsymbol{\Sigma}_1\mathbf{E}^\top(\mathbf{x} - \boldsymbol{\kappa})\right) + (\mathbf{x} - \boldsymbol{\kappa})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\kappa}),
\end{aligned}$$

where $\boldsymbol{\Sigma}_1^{-1} := \frac{1}{\sigma^2}\mathbf{I}_n + \frac{1}{\tau^2}\mathbf{E}^\top\mathbf{E}$. This implies that

$$\begin{aligned}
&\sum_{\mathbf{y}\in\mathbf{a}-\boldsymbol{\mu}+t\mathbb{Z}^n} \rho_{\sqrt{\boldsymbol{\Sigma}_1}}\left(\mathbf{y} - \frac{1}{\tau^2}\boldsymbol{\Sigma}_1\mathbf{E}^\top(\mathbf{x} - \boldsymbol{\kappa})\right) \cdot \rho_{\sqrt{\boldsymbol{\Sigma}}}(\mathbf{x} - \boldsymbol{\kappa}) \\
&= \rho_{\sqrt{\boldsymbol{\Sigma}_1}}\left(\mathbf{a} - \boldsymbol{\mu} - \frac{1}{\tau^2}\boldsymbol{\Sigma}_1\mathbf{E}^\top(\mathbf{x} - \boldsymbol{\kappa}) + t\mathbb{Z}^n\right) \cdot \rho_{\sqrt{\boldsymbol{\Sigma}}}(\mathbf{x} - \boldsymbol{\kappa}) \\
&\in \left[\frac{1-\epsilon}{1+\epsilon}, 1\right] \cdot \rho_{\sqrt{\boldsymbol{\Sigma}_1}}(t\mathbb{Z}^n) \cdot \rho_{\sqrt{\boldsymbol{\Sigma}}}(\mathbf{x} - \boldsymbol{\kappa}) \propto \left[\frac{1-\epsilon}{1+\epsilon}, 1\right] \cdot D_{\mathbb{Z}^m,\boldsymbol{\kappa},\sqrt{\boldsymbol{\Sigma}}}(\mathbf{x})
\end{aligned}$$

where the last inclusion is derived from Lem. 3 and the condition

$$s_{\min}(\sqrt{\mathbf{\Sigma}_1}) = \frac{1}{\left\|\mathbf{\Sigma}_1^{-1}\right\|} \geq \frac{1}{\sqrt{\frac{1}{\sigma^2} + \frac{1}{\tau^2}\left\|\mathbf{E}\right\|^2}} \geq \eta_\epsilon(t\mathbb{Z}^n).$$

As a result, we obtain

$$\mathcal{D}(\mathbf{x}) \in \left[\frac{1-\epsilon}{1+\epsilon}, \frac{1+\epsilon}{1-\epsilon}\right] \cdot D_{\mathbb{Z}^m, \boldsymbol{\kappa}, \sqrt{\mathbf{\Sigma}}}(\mathbf{x}) \subseteq [1-4\epsilon, 1+4\epsilon] \cdot D_{\mathbb{Z}^m, \boldsymbol{\kappa}, \sqrt{\mathbf{\Sigma}}}(\mathbf{x}),$$

and conclude that the statistical distance between $\mathcal{D}$ and $D_{\mathbb{Z}^m, \boldsymbol{\kappa}, \sqrt{\mathbf{\Sigma}}}$ is bounded by 4ϵ. □

Corollary 2. *Let $\mathbf{e} = (e_0, e_1, e_2)$ be an element of R^3 with $\|\mathbf{e}\|_\infty \leq B_\rho$. If*

$$\frac{1}{\sigma^2} + \frac{3n^2 B_\rho^2}{\tau^2} \leq \frac{1}{\eta_\epsilon(t\mathbb{Z}^n)^2}$$

for some $0 < \epsilon \leq 1/2$, then for any $\boldsymbol{\mu} \in R$ and $\boldsymbol{\nu} \in R^3$, the distribution of $\tilde{\mathbf{e}} = r \cdot \mathbf{e} + \mathbf{e}'$ over R^3, defined by $r \leftarrow D_{a+t\mathbb{Z}^n, \boldsymbol{\mu}, \sigma}$ and $\mathbf{e}' \leftarrow D_{\mathbb{Z}^{3n}, \boldsymbol{\nu}, \tau}$, is within statistical distance 4ϵ of $D_{\mathbb{Z}^{3n}, \boldsymbol{\kappa}, \sqrt{\mathbf{\Sigma}}}$ for $\boldsymbol{\kappa} = \boldsymbol{\mu} \cdot \mathbf{e} + \boldsymbol{\nu}$ and $\mathbf{\Sigma} = \sigma^2 \cdot \mathbf{E}\mathbf{E}^\top + \tau^2 \cdot \mathbf{I}_{3n}$, where $\mathbf{E}_i \in \mathbb{Z}^{n \times n}$ is the negacyclic matrix corresponding to e_i for $j = 0, 1, 2$ and $\mathbf{E} = \begin{bmatrix} \mathbf{E}_0 \\ \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix} \in \mathbb{Z}^{3n \times n}$.

Proof. Note that the coefficient representation of $r \cdot e_i$ can be written as $\mathbf{E}_i \mathbf{r}$, where $\mathbf{r}$ is the coefficient vector of r. Hence, this corollary is directly derived from the fact $\|\mathbf{E}\|^2 \leq 3n^2 B_\rho^2$ and Lem. 6. □

Theorem 1 (Linear Circuit Privacy)**.** *The BFV scheme, together with the randomized linear evaluation algorithm $\texttt{RandLinEval}(\mathbf{c}; a, b)$ under parameters σ and τ, is circuit-private for all affine functions over R_t if*

$$\frac{1}{\sigma^2} + \frac{3n^2 B_\rho^2}{\tau^2} \leq \frac{1}{\eta_\epsilon(t\mathbb{Z}^n)^2}$$

for some negligible $\epsilon > 0$.

Proof. Let $\mathbf{c} = \texttt{Enc}_{\mathbf{p}}(x, \mathbf{e})$ be an encryption of $x \in R_t$ for some $\mathbf{e} \leftarrow D^3_{\mathbb{Z}^n, \rho}$. Let $f(z) = az + b$ be an affine function over R_t for some coefficients $a, b \in R_t$, and $y = ax + b \in R_t$. We define the simulator Sim for the output ciphertext $\mathbf{c}'$ of our linear evaluation algorithm as $\texttt{RandLinEval}_{\mathbf{p}}(\mathbf{c}; 0, y)$, *i.e.,*

$$\mathsf{Sim}(s, \mathbf{p}, \mathbf{c}, y) = \{\mathbf{c}' = r \cdot \mathbf{c} + \texttt{Enc}_{\mathbf{p}}(y, \mathbf{e}') \in R_q^2 : r \leftarrow D_{t\mathbb{Z}^n, \sigma}, \mathbf{e}' \leftarrow D^3_{\mathbb{Z}^n, \tau}\}.$$

We use a hybrid argument to show that the distributions of $\mathbf{c}'$ from $\texttt{RandLinEval}(\mathbf{c}; a, b)$ and $\mathsf{Sim}(s, \mathbf{p}, \mathbf{c}, y)$ are statistically indistinguishable.

Let us consider the following distribution:

$$\mathcal{H} := \{\mathbf{c}' = \texttt{Enc}_{\mathbf{p}}(y, \tilde{\mathbf{e}}) : \tilde{\mathbf{e}} = (\tilde{e}_0, \tilde{e}_1, \tilde{e}_2) \leftarrow D_{\mathbb{Z}^{3n}, \sqrt{\mathbf{\Sigma}}}\}$$

where $\mathbf{\Sigma} = \sigma^2 \cdot \mathbf{E}\mathbf{E}^\top + \tau^2 \cdot \mathbf{I}_{3n}$ and $\mathbf{E}_i$ is the negacyclic matrix corresponding to e_i for $i = 0, 1, 2$ and $\mathbf{E} = \begin{bmatrix} \mathbf{E}_0 \\ \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix} \in \mathbb{Z}^{3n \times n}$.

We first claim that the real algorithms $\texttt{RandLinEval}$ and $\mathcal{H}$ are statistically indistinguishable for any a and b. As shown in the proof of Lem. 5, the output ciphertext from

$\texttt{RandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$ can be written as $\mathbf{c}' = \texttt{Enc}_{\mathbf{p}}(y, \tilde{\mathbf{e}})$ for some $r \in R$ sampled from $D_{a+t\mathbb{Z}^n,\sigma}$ and $\tilde{\mathbf{e}} = r\mathbf{e} + \mathbf{e}'$. Therefore, the distribution of $\texttt{RandLinEval}(\mathbf{c}; a, b)$ is within statistical distance 4ϵ of the distribution of $\mathcal{H}$ from Cor. 2.

Secondly, we claim that $\mathcal{H}$ and $\mathsf{Sim}(s, \mathbf{p}, \mathbf{c}, y)$ are also statistically indistinguishable. This can be shown by using the same argument as above, since $\mathsf{Sim}(s, \mathbf{p}, \mathbf{c}, y)$ is just a special case of $\texttt{RandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$ where $a = 0$ and $b = y$. Hence, the statistical distance between $\mathcal{H}$ and $\mathsf{Sim}(s, \mathbf{p}, \mathbf{c}, y)$ is bounded by 4ϵ.

Combining two claims, we conclude that $\texttt{RandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$ and $\mathsf{Sim}(s, \mathbf{p}, \mathbf{c}, y)$ are within a negligible statistical distance $\leq 8\epsilon$. □

It is worth noting that we do not use the secret key s when building the simulator in this theorem. In addition, the hybrid game $\mathcal{H}$ relies on the fact that $\mathbf{c}$ can be written in the form of $\texttt{Enc}_{\mathbf{p}}(x, \mathbf{e})$, and the randomness $\mathbf{e}$ is used to set the covariance matrix $\mathbf{\Sigma}$, but the simulator can be defined without this information.

Optimality. Now, we discuss the optimality of our algorithm from Thm. 1. Recall that the main condition on parameter selection is:

$$\frac{1}{\sigma^2} + \frac{3n^2 B_\rho^2}{\tau^2} \leq \frac{1}{\eta_\epsilon(t\mathbb{Z}^n)^2}.$$

Roughly speaking, the error parameters can be chosen so that σ and $\tau / \|\mathbf{E}\|$ have a small bitsize comparable to the smoothing parameter $\eta(t\mathbb{Z}^n)$. It is worth noting that these values can be bounded by $8t$ in practical parameter sets (see Table 1). As a result, the random variables $r \leftarrow D_{a+t\mathbb{Z}^n,\sigma}$ and $\mathbf{e}' \leftarrow D^3_{\mathbb{Z}^n,\tau}$ from the actual algorithm exhibit a comparable size to a and $t \cdot \mathbf{e}$ with only $\tilde{O}_\lambda(1)$ bit differences. Now, recall that the final noise of $\texttt{RandLinEval}$, denoted by $r \cdot \mathbf{e} + \mathbf{e}'$, possesses an asymptotic size of $a \cdot \mathbf{e}$, which corresponds to the final noise from $\texttt{LinEval}$, the linear evaluation without noise-flooding. Consequently, our linear evaluation algorithm can effectively manage asymptotically optimal sized parameters, resulting in only a few bit differences compared to the optimal case.

3.4 Randomized Linear Evaluation when $t \nmid q$

In this section, we discuss how to compute the randomized linear evaluation algorithm when the plaintext modulus t does not divide the ciphertext modulus q. This generalizes our OLE protocol to the BFV scheme with any parameters, as well as the BGV scheme [BGV12]. Roughly speaking, when $q = 1 \bmod t$, a BGV encryption of a message $x \in R_t$ has the form $(b, a) \in R_q$ where $b + as = x + te \bmod q$ for some small noise e. Then, it is easy to see that a BGV ciphertext can be converted into a BFV ciphertext by simply multiplying by a constant $(q-1)/t \in \mathbb{Z}$, where q and t denote the ciphertext and plaintext moduli, respectively. Then the affine function can be securely computed efficiently using our algorithm, and the resulting BFV ciphertext can be converted back to a BGV ciphertext by multiplying by a constant $-t$.

Now, let us discuss how to modify our randomized linear evaluation algorithm in order to operate with arbitrary parameters. Our key observation is that, given a ciphertext $\mathbf{c} \in R_q$ and the public key $\mathbf{p} \in R_q$, one can simply exploit our novel linear evaluation over R_{tq} instead of R_q.

Suppose that we are given a BFV encryption $\mathbf{c} = (c_0, c_1) \in R_q^2$ of $x \in R_t$. Observe that $t \cdot \mathbf{c} = (t \cdot c_0, t \cdot c_1) \in R_{tq}^2$ is a BFV encryption of x with a scaling factor q. Similarly, $t \cdot \mathbf{p} \in R_{tq}^2$ is a zero-encryption and thus can serve as a public key. Then, the new ciphertext modulus tq is a multiple of t, allowing our randomized linear evaluation algorithm to be applied to $t \cdot \mathbf{c}$ with public key $t \cdot \mathbf{p}$. Finally, the modulus can be switched back to q for consistency. Note that the resulting ciphertext of the linear evaluation is simulatable, and thus the final output is also simulatable. However, this naïve method requires modular

arithmetic over $\mathbb{Z}_{tq}$ instead of $\mathbb{Z}_q$. Therefore, we provide an optimized variant of this method, which operates over $\mathbb{Z}_q$.

• $\mathtt{ModiRandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$: Given a BFV ciphertext $\mathbf{c} \in R_q^2$ and ring elements $a, b \in R_t$, sample $r \leftarrow D_{a+t\mathbb{Z}^n,\sigma}$ and $(e'_0, e'_1, e'_2) \in R^3 \leftarrow D_{\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n},\tau}$. Compute and output $\mathbf{c}' = r \cdot \mathbf{c} + \mathtt{Enc}_{\mathbf{p}}(b, \tilde{\mathbf{e}}')$ where

$$\tilde{\mathbf{e}}' = \left(\left\lfloor (\frac{q}{t} - \left\lfloor \frac{q}{t} \right\rceil) \cdot b + e'_0 \right\rceil, e'_1, e'_2 \right) \in R^3.$$

Now we provide a correctness and security proof below.

Lemma 7 (Correctness)**.** *Let* $\mathbf{c} \leftarrow \mathtt{Enc}_{\mathbf{p}}(x)$ *for some* $x \in R_t$*. Then, the algorithm* $\mathtt{RandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$ *outputs a BFV encryption of* $f(x) = ax + b$ *if*

$$(nB_\rho B_\sigma + B_\tau)(1 + n + nB_\rho) + \frac{t+1}{2} < \frac{\Delta}{2}.$$

Proof. From the definition, we have $\mathbf{c} = \mathtt{Enc}_{\mathbf{p}}(x, \mathbf{e})$ and

$$\begin{aligned} \mathbf{c}' &= r \cdot \mathbf{c} + \mathtt{Enc}_{\mathbf{p}}(b, \mathbf{e}') \pmod q \\ &= \mathtt{Enc}_{\mathbf{p}}(y, r \cdot \mathbf{e} + \mathbf{e}') \pmod q \end{aligned}$$

where $\mathbf{e}' = (\lfloor (q/t - \lfloor q/t \rceil) \cdot b + e'_0 \rceil, e'_1, e'_2)$ for some $\mathbf{e} \leftarrow D^3_{\mathbb{Z}^n,\rho}$, $r \leftarrow D_{a+t\mathbb{Z}^n,\sigma}$ and $(e'_0, e'_1, e'_2) \leftarrow D_{\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n},\tau}$. Then, the only difference to $t \mid q$ case is $\lfloor (q/t - \lfloor q/t \rceil) \cdot b + e'_0 \rceil$ term, which is bounded by $1/2 \cdot t + B_\tau + 1/2 = B_\tau + (t+1)/2$. Therefore, it directly follows that the final noise is bounded by $\|e^*\|_\infty \leq (nB_\rho B_\sigma + B_\tau)(1 + n + nB_\rho) + (t+1)/2 < \Delta/2$ from the proof of Lem. 5, and the resulting ciphertext $\mathbf{c}'$ decrypts to $y = ax + b$ correctly. □

We remark that the additional $(t+1)/2$ term is almost negligible compared to the first term. To prove that our modified randomized linear evaluation algorithm achieves linear circuit privacy, we first prove a modified form of Corollary 2.

Lemma 8. *Let* $t > 0$ *be an integer,* $\mathbf{E} \in \mathbb{Z}^{3n \times n}$ *a matrix, and* $\sigma, \tau > 0$ *reals such that*

$$\frac{1}{\sigma^2} + \frac{1}{\tau^2} \|\mathbf{E}\|^2 \leq \frac{1}{\eta_\epsilon(t\mathbb{Z}^n)^2}$$

for some $0 < \epsilon \leq 1/2$*. Then, for arbitrary* $\mathbf{a}$*,* $\boldsymbol{\mu} \in \mathbb{R}^n$*, and* $\boldsymbol{\nu} \in \mathbb{R}^{3n}$*, the following distribution over* $\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n}$

$$\mathcal{D} := \left\{ \mathbf{E}\mathbf{r} + \mathbf{e}' : \mathbf{r} \leftarrow D_{\mathbf{a}+t\mathbb{Z}^n,\boldsymbol{\mu},\sigma}, \mathbf{e}' \leftarrow D_{\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n},\boldsymbol{\nu},\tau} \right\}$$

is within statistical distance 4ϵ *of* $D_{\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n},\boldsymbol{\kappa},\sqrt{\boldsymbol{\Sigma}}}$ *where* $\boldsymbol{\kappa} = \mathbf{E}\boldsymbol{\mu} + \boldsymbol{\nu}$ *and* $\boldsymbol{\Sigma} = \sigma^2 \cdot \mathbf{E}\mathbf{E}^\top + \tau^2 \cdot \mathbf{I}_{3n}$*.*

Proof. For $\mathbf{x} \in \mathbb{Z}^m$, the probability that $\mathcal{D}$ outputs $\mathbf{x}$ can be written as follows:

$$\begin{aligned} \mathcal{D}(\mathbf{x}) &= \Pr[\mathbf{E}\mathbf{r} + \mathbf{e}' = \mathbf{x} \mid \mathbf{r} \leftarrow D_{\mathbf{a}+t\mathbb{Z}^n,\boldsymbol{\mu},\sigma}, \mathbf{e}' \leftarrow D_{\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n},\boldsymbol{\nu},\tau}] \\ &= \sum_{\mathbf{y} \in \mathbf{a}+t\mathbb{Z}^n} D_{\mathbf{a}+t\mathbb{Z}^n,\boldsymbol{\mu},\sigma}(\mathbf{y}) \cdot D_{\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n},\boldsymbol{\nu},\tau}(\mathbf{x} - \mathbf{E}\mathbf{y}) \\ &\propto \sum_{\mathbf{y} \in \mathbf{a}+t\mathbb{Z}^n} \rho_\sigma(\mathbf{y} - \boldsymbol{\mu}) \cdot \rho_\tau(\mathbf{x} - \boldsymbol{\nu} - \mathbf{E}\mathbf{y}). \end{aligned}$$

Then, the rest of the proof directly follows from the proof of Lemma. 6. □

Theorem 2 (Linear Circuit Privacy)**.** *The modified randomized linear evaluation algorithm* $\texttt{ModiRandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$ *under parameters* σ *and* τ *is circuit-private for all affine functions over* R_t *if*

$$\frac{1}{\sigma^2} + \frac{3n^2 B_\rho^2}{\tau^2} \leq \frac{1}{\eta_\epsilon(t\mathbb{Z}^n)^2}$$

for some negligible $\epsilon > 0$*.*

Proof. Let $\mathbf{c} = \texttt{Enc}_{\mathbf{p}}(x, \mathbf{e})$ be an encryption of $x \in R_t$ for some $\mathbf{e} = (e_0, e_1, e_2) \leftarrow D^3_{\mathbb{Z}^n, \rho}$, and $f(z) = az + b$ be an affine function over R_t for some coefficients $a, b \in R_t$. Then, we define the simulator Sim for the output ciphertext $\mathbf{c}'$ of our linear evaluation algorithm as $\texttt{ModiRandLinEval}_{\mathbf{p}}(\mathbf{c}; 0, y)$ where $y = ax + b$. We use a hybrid argument to show that the distributions of $\mathbf{c}'$ from $\texttt{RandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$ and $\mathsf{Sim}(s, \mathbf{p}, \mathbf{c}, y)$ are statistically indistinguishable. First, observe that the distribution of $\texttt{ModiRandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$ is essentially identical to the following distribution.

$$\mathcal{H}_1 = \left\{ \mathbf{c}' = \left(\left\lfloor \frac{1}{t} c_0 \right\rceil, \left\lfloor \frac{1}{t} c_1 \right\rceil \right) \;\middle|\; \begin{array}{l} (e'_0, e'_1, e'_2) \in R^3 \leftarrow D_{\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n}, \tau}, r \leftarrow D_{a+t\mathbb{Z}^n, \sigma}, \\ (c_0, c_1) = t \cdot r \cdot \mathbf{c} + t \cdot \mathbf{p} \cdot e'_2 + (q \cdot b + t \cdot e'_0, t \cdot e'_1) \end{array} \right\}$$

This can be shown easily, since

$$\begin{aligned} \mathbf{c}' &= r \cdot \mathbf{c} + \mathbf{p} \cdot e'_2 + \left(\left\lfloor \frac{q}{t} \cdot b + e'_0 \right\rceil, e'_1 \right) \\ &= r \cdot \mathbf{c} + \mathbf{p} \cdot e'_2 + \left(\Delta b + \left\lfloor \left(\frac{q}{t} - \Delta \right) \cdot b + e'_0 \right\rceil, e'_1 \right) \\ &= r \cdot \mathbf{c} + \texttt{Enc}_{\mathbf{p}} \left(b; \mathbf{e}' = \left(\left\lfloor (\frac{q}{t} - \left\lfloor \frac{q}{t} \right\rceil) \cdot b + e'_0) \right\rceil, e'_1, e'_2 \right) \right). \end{aligned}$$

Now, consider the following distribution:

$$\mathcal{H}_2 := \left\{ \mathbf{c}' = \left(\left\lfloor \frac{1}{t} c_0 \right\rceil, \left\lfloor \frac{1}{t} c_1 \right\rceil \right) \;\middle|\; \begin{array}{l} (\tilde{e}_0, \tilde{e}_1, \tilde{e}_2) \in R^3 \leftarrow D_{\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n}, \sqrt{\mathbf{\Sigma}}}, \\ (c_0, c_1) = t \cdot \tilde{e}_2 \cdot \mathbf{p} + (q \cdot y + t \cdot \tilde{e}_0, t \cdot \tilde{e}_1) \end{array} \right\}$$

where $\mathbf{\Sigma} = \sigma^2 \cdot \mathbf{E}\mathbf{E}^\top + \tau^2 \cdot \mathbf{I}_{3n}$ and $\mathbf{E}_i$ is the negacyclic matrix corresponding to e_i for $i = 0, 1, 2$ and $\mathbf{E} = \begin{bmatrix} \mathbf{E}_0 \\ \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix} \in \mathbb{Z}^{3n \times n}$.

Then, by Lem. 8, $\mathcal{H}_1$ and $\mathcal{H}_2$ are within statistical distance 4ϵ. This can be shown easily with logic analogous to the proof of Lemma 1. Now, with an identical argument as above, we can show that the statistical distance of $\mathcal{H}_2$ and the following distribution $\mathcal{H}_3$ is bounded by 4ϵ since $\mathcal{H}_3$ is essentially a special case of $\mathcal{H}_1$ where $a = 0$ and $b = y$.

$$\mathcal{H}_3 = \left\{ \mathbf{c}' = \left(\left\lfloor \frac{1}{t} c_0 \right\rceil, \left\lfloor \frac{1}{t} c_1 \right\rceil \right) \;\middle|\; \begin{array}{l} (e'_0, e'_1, e'_2) \in R^3 \leftarrow D_{\frac{1}{t}\mathbb{Z}^n \times \mathbb{Z}^{2n}, \tau}, r \leftarrow D_{t\mathbb{Z}^n, \sigma}, \\ (c_0, c_1) = t \cdot r \cdot \mathbf{c} + t \cdot \mathbf{p} \cdot e'_2 + (q \cdot y + t \cdot e'_0, t \cdot e'_1) \end{array} \right\}$$

Finally, $\mathsf{Sim}(s, \mathbf{p}, \mathbf{c}, y)$ outputs an identical distribution as $\mathcal{H}_3$ as discussed above. To sum it all up, we conclude that the statistical distance between the two distributions $\texttt{ModiRandLinEval}_{\mathbf{p}}(\mathbf{c}; a, b)$ and $\mathsf{Sim}(s, \mathbf{p}, \mathbf{c}, y)$ is bounded by 8ϵ. □

4 Oblivious Linear Evaluation Protocols

In this section, we build efficient OLE protocols against passive adversaries by leveraging our circuit-private linear evaluation technique presented in the previous section.

Protocol Π_{OLE}

Receiver's input: A message $x \in R_t$

Sender's input: The coefficients $a, b \in R_t$ of a linear function $f(z) = az + b$

Setup Phase: The receiver generates a secret key $s \in R$ and a public key $\mathbf{p} = (p_0, p_1) \in R_q^2$, and sends $\mathbf{p}$ to the sender.

Encryption Phase: The receiver generates a ciphertext $\mathbf{c} \leftarrow \mathtt{Enc}_{\mathbf{p}}(x)$ and sends it to the sender.

Evaluation Phase: The sender performs $\mathbf{c}' = \mathtt{RandLinEval}_{\mathbf{p}}(\mathbf{c}, a, b)$ and sends it to the receiver.

Decryption Phase: The receiver outputs $y = \mathtt{Dec}_s(\mathbf{c}')$.

Figure 3: Semi-honest OLE protocol.

4.1 Secure OLE Protocol

Our OLE protocol in the semi-honest model Π_{OLE} is directly derived from the circuit-private evaluation technique in the previous section. Its formal description is given in Fig. 3.

Lemma 9 (Correctness)**.** *Suppose that both the receiver and the sender act honestly. Then, the protocol Π_{OLE} correctly computes the functionality $\mathcal{F}_{\mathsf{OLE}}$ with an overwhelming probability if $(nB_\rho B_\sigma + B_\tau)(1 + n + nB_\rho) < \Delta/2$.*

Proof. The proof of this lemma directly follows from Lem. 5. □

We now prove that Π_{OLE} (Fig. 3) securely computes $\mathcal{F}_{\mathsf{OLE}}$ (Fig. 2) in the presence of a static passive adversary.

Lemma 10 (Security Against Sender)**.** *The protocol Π_{OLE} satisfies receiver privacy against an honest-but-curious sender if the underlying BFV encryption scheme is semantically secure.*

Proof. We define $\mathcal{S}_{\mathsf{Receiver}}$ as a simulator that outputs a ciphertext $\mathbf{c} = \mathtt{Enc}_{\mathbf{p}}(0, \mathbf{e})$ with $\mathbf{e} \leftarrow D^3_{\mathbb{Z}^n,\rho}$, independent of the receiver's input x. Since the BFV encryption scheme is IND-CPA secure under the RLWE assumption of parameter (n, q, ρ), the simulated view from $\mathcal{S}_{\mathsf{Receiver}}$ is computationally indistinguishable from the sender's view in the real protocol Π_{OLE}. □

Lemma 11 (Security Against Receiver)**.** *The protocol Π_{OLE} is secure against an honest-but-curious receiver if*

$$\frac{1}{\sigma^2} + \frac{3n^2 B_\rho^2}{\tau^2} \leq \frac{1}{\eta_\epsilon(t\mathbb{Z}^n)^2} \tag{2}$$

for some negligible $\epsilon > 0$.

Proof. To show the security against the receiver, it suffices to show that the ciphertext $\mathbf{c}'$ is simulatable. The proof directly follows from Thm. 1. □

Theorem 3. *The protocol Π_{OLE} securely realizes the functionality $\mathcal{F}_{\mathsf{OLE}}$ in the presence of static passive adversaries in the $\mathcal{F}_{\mathsf{KeyGen}}$-hybrid model if the conditions of Lem. 9, 10, and 11 hold.*

Table 1: Recommended parameter sets for the semi-honest OLE with security level $\lambda = 128$.

$\log t$	n	$\log \sigma$	$\log \tau$	$\log q$
16	2^{12}	19	36	72
32	2^{12}	35	52	104
64	2^{13}	67	85	170
80	2^{13}	83	101	202
128	2^{14}	131	150	300

Table 2: Parameter Comparison for the OLE protocols.

Security	Scheme	$\log t$	$\log q$	n
Passive	[dCJV21]	32	220	2^{13}
	Ours		104	2^{12}

We remark that our scheme enjoys asymptotic optimality in terms of communication cost and practical efficiency. As discussed in Sec. 3.3, the underlying linear evaluation algorithm in our protocol achieves asymptotic optimality for the noise parameters, which extends to our OLE protocol as well. Therefore, our scheme can leverage a small ciphertext modulus, which only needs to accommodate the optimal-sized noise, thereby resulting in a low communication cost. We also present the recommended parameter sets, which support 16, 32, 64, 80, and 128 bits of the plaintext modulus for our semi-honest OLE protocol in Table 1. A more comprehensive analysis and comparison of the communication and computation costs of our protocol will be provided at the end of this section.

4.2 Performance Analysis

Here, we discuss the parameter selection for our improved OLE protocol, provide an analysis of its performance, and compare its performance to that of the prior art. We begin with a concise explanation of how the communication cost of the protocol is affected by the parameters. Recall that a bulk of communications during the protocol consists of elements in the polynomial ring $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, whose elements can be represented as a coefficient vector of size n, with each component bounded by q. Consequently, the communication cost for semi-malicious OLE is roughly proportional to $n \log q$. In order to minimize the communication cost of the scheme, we first fix the plaintext modulus t and then set q and n as small as possible.

4.2.1 Parameter Selection.

Parameter selection for our OLE protocol starts with the security parameter λ and the plaintext modulus t, both of which are provided by the application. The only restriction on the plaintext modulus is that $t = 1 \pmod{2n}$, where n is the degree of the polynomial modulus. This condition is necessary to support packing plaintext values into elements in the ring R_t, as explained in Section 2.4. The value of n is chosen to be the smallest power of 2 allowed by the ciphertext modulus; a larger ciphertext modulus requires a larger n to maintain security. Parameter selection begins with n as a small power of 2 (usually $n = 2^{12}$ at the start), and then, if the resulting ciphertext modulus is too large, the parameters are recomputed with a larger n. In practice, we never consider n beyond 2^{14}.

Given the plaintext modulus t and the polynomial modulus degree n, the next parameters to select are the Gaussian variances σ^2 and τ^2. Note that the initial error variance is set to be ρ^2, where ρ is fixed at 3.2 [ACC+21]. The remaining two variances must satisfy Inequality 2. Since the Gaussian sampling is the most computationally intensive part of the protocol, we minimize these variances by setting these two terms to be equal. In other words, we have $\sigma = \sqrt{2}\eta_\epsilon(t\mathbb{Z}^n)$ and $\tau = \sqrt{6}nB_\rho\eta_\epsilon(t\mathbb{Z}^n)$. Once we have these values, the only remaining parameter is the ciphertext modulus q. In order to guarantee correctness, we require that

$$(nB_\rho B_\sigma + B_\tau)(1 + n + nB_\rho) < \Delta/2 = q/2t.$$

If the resulting bound on q is larger than that allowed by the current value of n, then we restart the parameter selection process with $n \leftarrow 2n$. We make use of the lattice estimator [APS15] to verify the largest q associated with n.

4.2.2 Performance.

In [dCJV21], the authors present a secure OLE protocol against semi-honest adversaries, providing 80-bit security. In their parameter set, $\log t \approx 32, \log q \approx 220$ and $n = 2^{13}$. On the other hand, in our construction, we can leverage a smaller ciphertext modulus and the ring dimension of $\log q \approx 104$ and $n = 2^{12}$ with the same size of plaintext modulus t, under 128-bit security.

Table 3: OLE performance comparison of our protocol and the protocol from [dCJV21]

	$\log t$	n	Elapsed Time	Communication Cost	Estimated Communication time	Time per OLE	Bit-Security
Ours	32	2^{12}	4.25ms	256.6KiB	20.0ms	5.92μs	128-bit
[dCJV21]	32	2^{13}	4.89ms	768.6KiB	60.0ms	7.92μs	80-bit

We also provide a proof-of-concept implementation of the semi-honest version of our protocol and compare it to the previous work [dCJV21]. The proposed method and the prior method [dCJV21] were implemented in Lattigo v5 [MBTPH20], and all the experiments were performed on a machine with an Intel(R) Xeon(R) Platinum 8268 @ 2.90GHz CPU and 192GB RAM running Ubuntu 20.04.2 LTS. We estimate the communication time, assuming that the parties are connected by a network with a bandwidth of 100Mbps (WAN). Table. 3 describes the benchmark results. Our protocol achieves a ciphertext size that is 3 times more compact than the construction from [dCJV21], as discussed above. Moreover, the amortized time of our protocol improves upon prior work by 33%. It is worth noting that the main bottleneck of our method is Gaussian sampling, while the rescaling operation is also a time-consuming operation in the construction of [dCJV21]. In fact, this can bring much more efficiency in the real-world setting since the sender can pre-compute Gaussian samples during the communication, while rescaling cannot be performed until the sender receives the ciphertext from the receiver. In our implementation, a rounded Gaussian Sampler was utilized instead of discrete Gaussian for sampling from $D^3_{\mathbb{Z}^n,\tau}$. Note that the statistical distance between the rounded Gaussian and discrete Gaussian is small enough since τ is sufficiently large in our parameters (≥ 36 bits).

5 Discussion

Semi-malicious security. We highlight that our linear-evaluation algorithm achieves more than traditional circuit privacy. In the standard definition of circuit privacy, we assume a semi-honest receiver who honestly samples randomness for the secret and public key pair, as well as for ciphertexts. This definition cannot be easily extended to the active setting with a fully-malicious receiver, even in the presence of zero-knowledge proofs of knowledge (ZKPoK), since we need to prove not only the well-formedness of the public key and ciphertexts but also the correctness of the distribution of the randomness. However, ZKPoKs can still reduce the adversarial model from fully malicious to "semi-malicious", meaning the adversary follows the protocol but may choose randomness adversarially.

Interestingly, our randomized linear evaluation algorithm ensures the privacy of the sender's input even against a semi-malicious receiver. Recall that Thm. 1 implies that the output ciphertext of our randomized linear evaluation algorithm can be simulated as $r \cdot \mathbf{c} + \mathtt{Enc}_{\mathbf{p}}(y, \mathbf{e}')$ where r and $\mathbf{e}'$ are drawn from some discrete Gaussian distributions independent from a and b. This simulation does not rely on any structural aspects of the public key $\mathbf{p}$. Hence, the receiver learns no information about a or b, even if the randomness used for generating sk, pk, or ciphertexts is adversarially chosen. When combined with ZKPoKs, this yields full malicious security for our protocol. In the full version of the paper, we formally describe how to obtain full malicious security.

References

[ACC+21] Martin Albrecht, Melissa Chase, Hao Chen, Jintai Ding, Shafi Goldwasser, Sergey Gorbunov, Shai Halevi, Jeffrey Hoffstein, Kim Laine, Kristin Lauter, et al. Homomorphic encryption standard. In *Protecting Privacy through Homomorphic Encryption*, pages 31–62. Springer, 2021. doi:10.1007/978-3-030-77287-1_2.

[ADI+17] Benny Applebaum, Ivan Damgård, Yuval Ishai, Michael Nielsen, and Lior Zichron. Secure arithmetic computation with constant computational overhead. In Jonathan Katz and Hovav Shacham, editors, *Advances in Cryptology – CRYPTO 2017, Part I*, volume 10401 of *Lecture Notes in Computer Science*, pages 223–254, Santa Barbara, CA, USA, August 20–24, 2017. Springer, Cham, Switzerland. doi:10.1007/978-3-319-63688-7_8.

[AGHS13] Shweta Agrawal, Craig Gentry, Shai Halevi, and Amit Sahai. Discrete Gaussian leftover hash lemma over infinite domains. In Kazue Sako and Palash Sarkar, editors, *Advances in Cryptology – ASIACRYPT 2013, Part I*, volume 8269 of *Lecture Notes in Computer Science*, pages 97–116, Bengalore, India, December 1–5, 2013. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-642-42033-7_6.

[APS15] Martin Albrecht, Rachel Player, and Sam Scott. On the concrete hardness of learning with errors. *Journal of Mathematical Cryptology*, 9, 10 2015. doi:10.1515/jmc-2015-0016.

[BBDP22] Zvika Brakerski, Pedro Branco, Nico Döttling, and Sihang Pu. Batch-OT with optimal rate. In Orr Dunkelman and Stefan Dziembowski, editors, *Advances in Cryptology – EUROCRYPT 2022, Part II*, volume 13276 of *Lecture Notes in Computer Science*, pages 157–186, Trondheim, Norway, May 30 – June 3, 2022. Springer, Cham, Switzerland. doi:10.1007/978-3-031-07085-3_6.

[BCG+20] Elette Boyle, Geoffroy Couteau, Niv Gilboa, Yuval Ishai, Lisa Kohl, and Peter Scholl. Efficient pseudorandom correlation generators from ring-LPN. In Daniele Micciancio and Thomas Ristenpart, editors, *Advances in Cryptology – CRYPTO 2020, Part II*, volume 12171 of *Lecture Notes in Computer Science*, pages 387–416, Santa Barbara, CA, USA, August 17–21, 2020. Springer, Cham, Switzerland. doi:10.1007/978-3-030-56880-1_14.

[BCGI18] Elette Boyle, Geoffroy Couteau, Niv Gilboa, and Yuval Ishai. Compressing vector OLE. In David Lie, Mohammad Mannan, Michael Backes, and XiaoFeng Wang, editors, *ACM CCS 2018: 25th Conference on Computer and Communications Security*, pages 896–912, Toronto, ON, Canada, October 15–19, 2018. ACM Press. doi:10.1145/3243734.3243868.

[BDM22] Pedro Branco, Nico Döttling, and Paulo Mateus. Two-round oblivious linear evaluation from learning with errors. In Goichiro Hanaoka, Junji Shikata, and Yohei Watanabe, editors, *PKC 2022: 25th International Conference on Theory and Practice of Public Key Cryptography, Part I*, volume 13177 of *Lecture Notes in Computer Science*, pages 379–408, Virtual Event, March 8–11, 2022. Springer, Cham, Switzerland. doi:10.1007/978-3-030-97121-2_14.

[BDOZ11] Rikke Bendlin, Ivan Damgård, Claudio Orlandi, and Sarah Zakarias. Semi-homomorphic encryption and multiparty computation. In Kenneth G. Paterson, editor, *Advances in Cryptology – EUROCRYPT 2011*, volume 6632 of *Lecture Notes in Computer Science*, pages 169–188, Tallinn, Estonia, May 15–19, 2011. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-642-20465-4_11.

[BdPMW16] Florian Bourse, Rafaël del Pino, Michele Minelli, and Hoeteck Wee. FHE circuit privacy almost for free. In Matthew Robshaw and Jonathan Katz, editors, *Advances in Cryptology – CRYPTO 2016, Part II*, volume 9815 of *Lecture Notes in Computer Science*, pages 62–89, Santa Barbara, CA, USA, August 14–18, 2016. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-662-53008-5_3.

[BEPU+20] Carsten Baum, Daniel Escudero, Alberto Pedrouzo-Ulloa, Peter Scholl, and Juan Ramón Troncoso-Pastoriza. Efficient protocols for oblivious linear function evaluation from ring-LWE. In Clemente Galdi and Vladimir Kolesnikov, editors, *SCN 20: 12th International Conference on Security in Communication Networks*, volume 12238 of *Lecture Notes in Computer Science*, pages 130–149, Amalfi, Italy, September 14–16, 2020. Springer, Cham, Switzerland. doi:10.1007/978-3-030-57990-6_7.

[BGMN18] Alexander R. Block, Divya Gupta, Hemanta K. Maji, and Hai H. Nguyen. Secure computation using leaky correlations (asymptotically optimal constructions). In Amos Beimel and Stefan Dziembowski, editors, *TCC 2018: 16th Theory of Cryptography Conference, Part II*, volume 11240 of *Lecture Notes in Computer Science*, pages 36–65, Panaji, India, November 11–14, 2018. Springer, Cham, Switzerland. doi:10.1007/978-3-030-03810-6_2.

[BGV12] Zvika Brakerski, Craig Gentry, and Vinod Vaikuntanathan. (Leveled) fully homomorphic encryption without bootstrapping. In Shafi Goldwasser, editor, *ITCS 2012: 3rd Innovations in Theoretical Computer Science*, pages 309–325, Cambridge, MA, USA, January 8–10, 2012. Association for Computing Machinery. doi:10.1145/2090236.2090262.

[Bra12] Zvika Brakerski. Fully homomorphic encryption without modulus switching from classical GapSVP. In Reihaneh Safavi-Naini and Ran Canetti, editors, *Advances in Cryptology – CRYPTO 2012*, volume 7417 of *Lecture Notes in Computer Science*, pages 868–886, Santa Barbara, CA, USA, August 19–23, 2012. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-642-32009-5_50.

[CDI+19] Melissa Chase, Yevgeniy Dodis, Yuval Ishai, Daniel Kraschewski, Tianren Liu, Rafail Ostrovsky, and Vinod Vaikuntanathan. Reusable non-interactive secure computation. In Alexandra Boldyreva and Daniele Micciancio, editors, *Advances in Cryptology – CRYPTO 2019, Part III*, volume 11694 of *Lecture Notes in Computer Science*, pages 462–488, Santa Barbara, CA, USA, August 18–22, 2019. Springer, Cham, Switzerland. doi:10.1007/978-3-030-26954-8_15.

[CGS97] Ronald Cramer, Rosario Gennaro, and Berry Schoenmakers. A secure and optimally efficient multi-authority election scheme. In Walter Fumy, editor, *Advances in Cryptology – EUROCRYPT'97*, volume 1233 of *Lecture Notes in Computer Science*, pages 103–118, Konstanz, Germany, May 11–15, 1997. Springer Berlin Heidelberg, Germany. doi:10.1007/3-540-69053-0_9.

[CILO22] Wutichai Chongchitmate, Yuval Ishai, Steve Lu, and Rafail Ostrovsky. PSI from ring-OLE. In Heng Yin, Angelos Stavrou, Cas Cremers, and Elaine Shi, editors, *ACM CCS 2022: 29th Conference on Computer and Communications Security*, pages 531–545, Los Angeles, CA, USA, November 7–11, 2022. ACM Press. doi:10.1145/3548606.3559378.

[CRR21] Geoffroy Couteau, Peter Rindal, and Srinivasan Raghuraman. Silver: Silent VOLE and oblivious transfer from hardness of decoding structured LDPC codes. In Tal Malkin and Chris Peikert, editors, *Advances in Cryptology – CRYPTO 2021, Part III*, volume 12827 of *Lecture Notes in Computer Science*, pages 502–534, Virtual Event, August 16–20, 2021. Springer, Cham, Switzerland. doi:10.1007/978-3-030-84252-9_17.

[dCHI+22] Leo de Castro, Carmit Hazay, Yuval Ishai, Vinod Vaikuntanathan, and Muthu Venkitasubramaniam. Asymptotically quasi-optimal cryptography. In Orr Dunkelman and Stefan Dziembowski, editors, *Advances in Cryptology – EUROCRYPT 2022, Part I*, volume 13275 of *Lecture Notes in Computer Science*, pages 303–334, Trondheim, Norway, May 30 – June 3, 2022. Springer, Cham, Switzerland. doi:10.1007/978-3-031-06944-4_11.

[dCJV21] Leo de Castro, Chiraag Juvekar, and Vinod Vaikuntanathan. Fast vector oblivious linear evaluation from ring learning with errors. In *Proceedings of the 9th on Workshop on Encrypted Computing & Applied Homomorphic Cryptography*, pages 29–41, 2021. doi:10.1145/3474366.3486928.

[DGN+17] Nico Döttling, Satrajit Ghosh, Jesper Buus Nielsen, Tobias Nilges, and Roberto Trifiletti. TinyOLE: Efficient actively secure two-party computation from oblivious linear function evaluation. In Bhavani M. Thuraisingham, David Evans, Tal Malkin, and Dongyan Xu, editors, *ACM CCS 2017: 24th Conference on Computer and Communications Security*, pages 2263–2276, Dallas, TX, USA, October 31 – November 2, 2017. ACM Press. doi:10.1145/3133956.3134024.

[DS16] Léo Ducas and Damien Stehlé. Sanitization of FHE ciphertexts. In Marc Fischlin and Jean-Sébastien Coron, editors, *Advances in Cryptology – EUROCRYPT 2016, Part I*, volume 9665 of *Lecture Notes in Computer Science*, pages 294–310, Vienna, Austria, May 8–12, 2016. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-662-49890-3_12.

[EGP+23] Daniel Escudero, Vipul Goyal, Antigoni Polychroniadou, Yifan Song, and Chenkai Weng. SuperPack: Dishonest majority MPC with constant online communication. In Carmit Hazay and Martijn Stam, editors, *Advances in Cryptology – EUROCRYPT 2023, Part II*, volume 14005 of *Lecture Notes in Computer Science*, pages 220–250, Lyon, France, April 23–27, 2023. Springer, Cham, Switzerland. doi:10.1007/978-3-031-30617-4_8.

[FV12] Junfeng Fan and Frederik Vercauteren. Somewhat practical fully homomorphic encryption. Cryptology ePrint Archive, Report 2012/144, 2012. URL: https://eprint.iacr.org/2012/144.

[Gen09] Craig Gentry. Fully homomorphic encryption using ideal lattices. In Michael Mitzenmacher, editor, *41st Annual ACM Symposium on Theory of Computing*, pages 169–178, Bethesda, MD, USA, May 31 – June 2, 2009. ACM Press. doi:10.1145/1536414.1536440.

[GN19] Satrajit Ghosh and Tobias Nilges. An algebraic approach to maliciously secure private set intersection. In Yuval Ishai and Vincent Rijmen, editors, *Advances in Cryptology – EUROCRYPT 2019, Part III*, volume 11478 of *Lecture Notes in Computer Science*, pages 154–185, Darmstadt, Germany, May 19–23, 2019. Springer, Cham, Switzerland. doi:10.1007/978-3-030-17659-4_6.

[GPV08] Craig Gentry, Chris Peikert, and Vinod Vaikuntanathan. Trapdoors for hard lattices and new cryptographic constructions. In Richard E. Ladner and Cynthia Dwork, editors, *40th Annual ACM Symposium on Theory of Computing*, pages 197–206, Victoria, BC, Canada, May 17–20, 2008. ACM Press. doi:10.1145/1374376.1374407.

[GSW13] Craig Gentry, Amit Sahai, and Brent Waters. Homomorphic encryption from learning with errors: Conceptually-simpler, asymptotically-faster, attribute-based. In Ran Canetti and Juan A. Garay, editors, *Advances in Cryptology – CRYPTO 2013, Part I*, volume 8042 of *Lecture Notes in Computer Science*, pages 75–92, Santa Barbara, CA, USA, August 18–22, 2013. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-642-40041-4_5.

[HIMV19] Carmit Hazay, Yuval Ishai, Antonio Marcedone, and Muthuramakrishnan Venkitasubramaniam. LevioSA: Lightweight secure arithmetic computation. In Lorenzo Cavallaro, Johannes Kinder, XiaoFeng Wang, and Jonathan Katz, editors, *ACM CCS 2019: 26th Conference on Computer and Communications Security*, pages 327–344, London, UK, November 11–15, 2019. ACM Press. doi:10.1145/3319535.3354258.

[IP07] Yuval Ishai and Anat Paskin. Evaluating branching programs on encrypted data. In Salil P. Vadhan, editor, *TCC 2007: 4th Theory of Cryptography Conference*, volume 4392 of *Lecture Notes in Computer Science*, pages 575–594, Amsterdam, The Netherlands, February 21–24, 2007. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-540-70936-7_31.

[IPS09] Yuval Ishai, Manoj Prabhakaran, and Amit Sahai. Secure arithmetic computation with no honest majority. In Omer Reingold, editor, *TCC 2009:*

6th Theory of Cryptography Conference, volume 5444 of *Lecture Notes in Computer Science*, pages 294–314. Springer Berlin Heidelberg, Germany, March 15–17, 2009. doi:10.1007/978-3-642-00457-5_18.

[KPR18] Marcel Keller, Valerio Pastro, and Dragos Rotaru. Overdrive: Making SPDZ great again. In Jesper Buus Nielsen and Vincent Rijmen, editors, *Advances in Cryptology – EUROCRYPT 2018, Part III*, volume 10822 of *Lecture Notes in Computer Science*, pages 158–189, Tel Aviv, Israel, April 29 – May 3, 2018. Springer, Cham, Switzerland. doi:10.1007/978-3-319-78372-7_6.

[MBTPH20] Christian Vincent Mouchet, Jean-Philippe Bossuat, Juan Ramón Troncoso-Pastoriza, and Jean-Pierre Hubaux. Lattigo: A multiparty homomorphic encryption library in go. In *Proceedings of the 8th Workshop on Encrypted Computing and Applied Homomorphic Cryptography*, pages 64–70, 2020. doi:10.25835/0072999.

[MR07] Daniele Micciancio and Oded Regev. Worst-case to average-case reductions based on gaussian measures. *SIAM Journal on Computing*, 37(1):267–302, 2007. doi:10.1137/S0097539705447360.

[NP99] Moni Naor and Benny Pinkas. Oblivious transfer and polynomial evaluation. In *31st Annual ACM Symposium on Theory of Computing*, pages 245–254, Atlanta, GA, USA, May 1–4, 1999. ACM Press. doi:10.1145/301250.301312.

[OS07] Rafail Ostrovsky and William E. Skeith, III. A survey of single-database private information retrieval: Techniques and applications (invited talk). In Tatsuaki Okamoto and Xiaoyun Wang, editors, *PKC 2007: 10th International Conference on Theory and Practice of Public Key Cryptography*, volume 4450 of *Lecture Notes in Computer Science*, pages 393–411, Beijing, China, April 16–20, 2007. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-540-71677-8_26.

[Pei10] Chris Peikert. An efficient and parallel Gaussian sampler for lattices. In Tal Rabin, editor, *Advances in Cryptology – CRYPTO 2010*, volume 6223 of *Lecture Notes in Computer Science*, pages 80–97, Santa Barbara, CA, USA, August 15–19, 2010. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-642-14623-7_5.

[RAD78] Ronald L Rivest, Len Adleman, and Michael L Dertouzos. On data banks and privacy homomorphisms. *Foundations of secure computation*, 4(11):169–180, 1978.

[RS21] Peter Rindal and Phillipp Schoppmann. VOLE-PSI: Fast OPRF and circuit-PSI from vector-OLE. In Anne Canteaut and François-Xavier Standaert, editors, *Advances in Cryptology – EUROCRYPT 2021, Part II*, volume 12697 of *Lecture Notes in Computer Science*, pages 901–930, Zagreb, Croatia, October 17–21, 2021. Springer, Cham, Switzerland. doi:10.1007/978-3-030-77886-6_31.

[WYKW21] Chenkai Weng, Kang Yang, Jonathan Katz, and Xiao Wang. Wolverine: Fast, scalable, and communication-efficient zero-knowledge proofs for boolean and arithmetic circuits. In *2021 IEEE Symposium on Security and Privacy*, pages 1074–1091, San Francisco, CA, USA, May 24–27, 2021. IEEE Computer Society Press. doi:10.1109/SP40001.2021.00056.

[YSWW21] Kang Yang, Pratik Sarkar, Chenkai Weng, and Xiao Wang. QuickSilver: Efficient and affordable zero-knowledge proofs for circuits and polynomials over any field. In Giovanni Vigna and Elaine Shi, editors, *ACM CCS 2021: 28th Conference on Computer and Communications Security*, pages 2986–3001, Virtual Event, Republic of Korea, November 15–19, 2021. ACM Press. doi:10.1145/3460120.3484556.